ASHMILA ASHRAF

SOFTWARE ENGINEER

ashmilaashrafofficial@gmail.com | +919074283539 | linkedin.com/in/ashmila-ashraf

PROFESSIONAL SUMMARY

Software Engineer with 2.5+ years of experience developing secure web applications for national-level government systems. Specialized in application security implementation, real-time monitoring systems, and end-to-end development using **ReactJS**, **Spring Boot**, **Node.js** and **Python**. Proven expertise in **security vulnerability remediation** following OWASP WSTG standards, **web application security testing**, API integration and secure authentication implementation.

EXPERIENCE

Project Engineer **Jun 2023 - Feb 2026**
Centre for Development of Advanced Computing, India *Thiruvananthapuram, KL, IN*

- Independently architect, develop, and deploy **Full-stack web applications** for national projects, managing complete lifecycle from requirements to production deployment
- **Application Security:** Remediated critical vulnerabilities post internal penetration testing includes authentication & authorization, credential encryption, server-side session management, IP-based rate limiting with account lockout, error handling, security headers configuration, input validation & sanitization, and XSS/SQL injection prevention
- Developed secure authentication using Keycloak SSO with HttpOnly cookies, RSA/ AES encryption, API-key authentication, RBAC and environment-based secrets management
- Build RESTful APIs for data management, real-time updates, and frontend integration; integrate multiple internal department APIs for complex data visualization with geospatial mapping (React Leaflet)
- Managed application deployment; developed shell scripts for deployments; configure NGINX, WildFly, and Syslog-ng centralized logging infrastructure on RHEL 9.4
- **VAPT :** Application/API risk assessment, secure code reviews, and web app security testing using Burp Suite, Snyk (SAST/SCA), Postman and manual testing methodologies per OWASP WSTG standards

PROJECTS

ERSS Scenario Replay (CDAC) **Jul 2023 - Sept 2024**

- ***Technology:*** *ReactJS 18, NodeJS, React Chrono, React Leaflet, Material UI, Keycloak, NGINX*
- Independently developed comprehensive timeline visualization application for end-to-end emergency event data representation with multimedia integration, nested timelines for multi-vehicle dispatch tracking, geospatial mapping with vehicle route visualization and PDF report generation
- Implemented secure authentication - AES & Keycloak SSO
- Deployed across Phase 1 and Phase 2 emergency response support system (ERSS) states

ERSS Alarms and Alerts (CDAC) **May 2024 - Oct 2025**

- ***Technology:*** *ReactJS 17, Spring Boot 2.6, Java 11, Python 3, PostgreSQL, Syslog-ng, Wildfly 21*
- Transformed partially developed system into production-ready, multi-state real-time log monitoring platform with centralized national dashboard featuring interactive India map
- Built full-stack solution: ReactJS frontend, Spring Boot backend with RESTful APIs, and Python scripts for automated log processing via Syslog-ng data pipeline
- Centralized Logging System: Set-up Syslog-ng server configuration for capturing logs from diverse technology stack (Java, Node.js, Python applications)
- Implemented comprehensive security hardening post-audit

ERSS Call Center Performance Monitoring System (CDAC) **Mar 2025 - Dec 2025**

- ***Technologies:*** *ReactJS 19 (Vite), NodeJS, Keycloak, Material-UI, Radix UI, Shadcn UI, Lucid react*
- Developed a real-time performance monitoring web application for tracking emergency response call taker centers with live agent availability tracking, threshold-based alerts, real-time status indicators, and interactive analytics dashboards
- Built comprehensive report pages with filters and exportable analytics; integrated REST APIs of internal reporting team and implemented Keycloak role-based authentication

Additional Contributions

- Managed maintenance of ERSS Call Tracing application: handling bug resolution, troubleshooting, and deployment
- Developed standardized Syslog-ng configuration templates for Java, Node.js and Python based applications, adopted by multiple development teams
- Prepared technical documentation including architecture diagrams, API documentation, and deployment procedures

ACADEMIC PROJECT

SignQuiz **2020**

- ***Technology:*** *Python, Django, TensorFlow, Keras, OpenCV*
- Developed a Python and Django-based based quiz web application for learning Indian Sign Language fingerspelled signs using CNN for image classification
- Implemented webcam-based image capture with OpenCV and trained a basic CNN model for sign recognition

SKILLS

- **Programming Languages:** Python, Java
- **Frontend Technologies:** ReactJS (v17, v18, v19 - Vite), HTML5, CSS3
- **Backend Frameworks:** Node.js, Spring Boot
- **React Ecosystem:** Material-UI, React Bootstrap, React Chrono, React Leaflet, React Redux
- **Database:** PostgreSQL
- **Security Practices / Management:** Secure Code Review, Risk Assessment, Keycloak, API-key
- **Security & Testing Tools:** Burp Suite, Snyk, Postman, Bandit
- **API Development:** RESTful APIs, API Integration
- **DevOps / Tools & Platforms:** Git, Github, Nginx, WildFly, RHEL, PgAdmin, Syslog-ng, AWS
- **AI Tools:** Claude ai, v0 by Vercel, Lovable, Chatgpt, Anti-gravity, Gemini, Deepseek, Grok
- **Other:** Problem Solving, Manual Debugging, Security Vulnerability Assessment & Remediation

ACHIEVEMENTS

- **Copyright Registration:** ERSS Scenario Replay web application Registered under the Indian Copyright Act, 1957 (Certificate No. SW-2025020784)

EDUCATION

Bachelor of Technology in Computer Science and Engineering **2016-2020**
Malabar Institute of Technology - APJ Abdul Kalam Technological University
CGPA: 8.79

LANGUAGES

- English, Malayalam